

Arsalan khan

Cybersecurity Student | AI-Driven Network Security | Threat Detection

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📍 Islamabad, Pakistan, ZIP Code: 44000

SUMMARY

A final-year Cybersecurity student at GIKI with a passion for understanding how attackers operate at the network level and building AI-driven systems to stop them. My work spans malware detection, network traffic analysis, and intrusion detection grounded in real-world datasets and threat models. I bring both technical depth and hands-on leadership experience, and I am now seeking a graduate research opportunity where I can push this work further in cybersecurity risk management and APT detection.

EXPERIENCE

Pakistan Air Force

Jul 2025 – Aug 2025

Cyber Security Intern

Islamabad, Pakistan

- Assisted in SOC operations and network security configurations, including vulnerability assessments and penetration testing on lab environments.
- Performed end-to-end security testing using Burp Suite and labs to identify and mitigate web-based vulnerabilities.

ITSOLERA PVT LTD

Jul 2024 - Sept 2024

Red Team Intern

Remote

- Conducted penetration testing and threat analysis to identify weaknesses and improve overall security posture.
- Developed "ConnectFinder," a Python-based OSINT tool designed to aggregate data from multiple sources for identity verification and threat research.

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Sept 2022 - Jun 2026

Bachelors of Science in Cyber Security

Topi, Pakistan

- CGPA:** 3.43/4.00 | **Final Year Project:** TrustChain: A Blockchain-based Secure Voting System

PROJECTS

Malware Traffic Detection Using Victim-Attacker Interaction Patterns

Jan 2026

Tools: Python, PyTorch, Scapy, LightGBM, Google Colab

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- Built an ML-based system to detect and classify 10 malware families from encrypted network traffic by engineering 22 statistical features from Victim-Attacker interaction patterns, achieving 99.88% binary detection accuracy and 96.24% multi-class classification accuracy.

Network Forensic Framework for APT Detection

Oct 2025

Tools: Python, MLP Deep Neural Network, TensorFlow/Keras, UNSW-NB15

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- Designed an APT detection framework using a fine-tuned MLP Deep Neural Network on the UNSW-NB15 dataset, with rigorous feature selection across 49 network attributes to catch subtle, multi-stage network intrusions.

Anomaly-Based Intrusion Detection System (IDS)

Jul 2025

Tools: Python, K-Means Clustering, PCA, Scikit-Learn

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- Built an unsupervised IDS using K-Means Clustering and PCA across 40+ traffic features to profile normal behavior and surface zero-day threats, achieving a Silhouette Score of 0.87 confirming high-confidence anomaly separation.

AWARDS

Dean's Honor Roll Award — Achieved distinction GPA for last five consecutive semesters.

LEADERSHIP EXPERIENCE

Member & Hackathon Head – ACM GIKI Chapter

Sept 2023 – Present

Ghulam Ishaq Khan Institute (GIKI)

- Orchestrated technical competitions and hackathons for 250+ participants, managing the full event lifecycle from logistics to execution.

Head Member – SOPHEP

Feb 2023 – Present

Ghulam Ishaq Khan Institute (GIKI)

- Led the organization of flagship institutional events and career fairs, coordinating with 50+ industry leaders to facilitate recruitment.

SKILLS

Network Security & Threat Detection: Network Traffic/Malware Analysis, Suricata IDS, Wireshark, SIEM

Machine Learning & AI: LightGBM, MLP/DNN, TensorFlow, PyTorch, K-Means, Scikit-Learn

Programming: Python, C++, SQL, Bash, Flask, HTML/CSS/JavaScript, REST APIs

Penetration Testing & Red Teaming: Metasploit, Burp Suite, Kali Linux, OWASP, Vulnerability Assessment

Cloud & DevOps: AWS, Terraform, Docker, GitHub Actions (CI/CD), Prometheus, Grafana

ADDITIONAL INFORMATION

Languages: English (Fluent), Urdu (Fluent), Pashto (Native)

Interests: SOC Operations, Network Defense, Honeynet, Football, Music, Traveling